MedBuddy System Architecture: The Engine Room

User Interface & Interaction

Streamlit



Streamlit Web App: Seamless, real-time query input and response rendering.

API Requests / Chat Responses

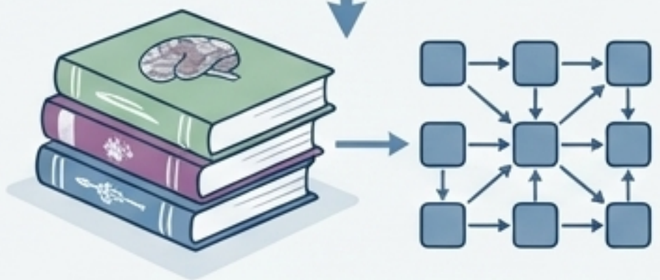
RAG Pipeline & Data Management

LlamaIndex Framework: Document indexing, vector storage, and semantic retrieval.

Context Ingestion & Prompt Execution

Generative Engine

Gemini LLM: High-level reasoning, medical context synthesis, and response generation.



Raw Medical PDFs → Vector Embeddings.

Grounding & Safety

TruLens Evaluation: Hallucination checking and 84% accuracy validation against source medical texts.

Grounding Verification

NotebookLM

The RAG Workflow: From Raw Text to Verified Medical Insight

Phase 1: Ingestion (Offline Process)



Document Upload



Parsing & Chunking
LlamaIndex Parsing



Embedding
Text-to-Vector Conversion

Phase 2: Querying (Live Interaction)



Vector Database
LlamaIndex Vector Store



Semantic Search
LlamaIndex Similarity Search



User Query



Prompt Augmentation

Gemini

Generation
LLM Synthesizes Answer from Context Only

TruLens

Evaluation & Delivery
Hallucination Check

NotebookLM